

# Log Summary

-----  
Warmstart time: 0.43 min

BDE time: 39.37 min

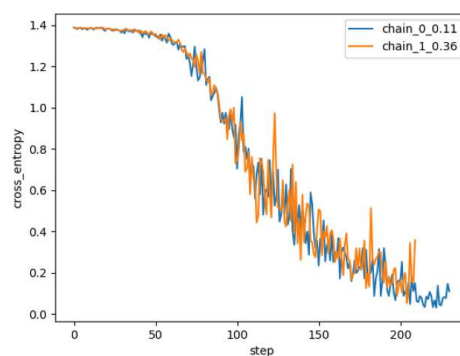
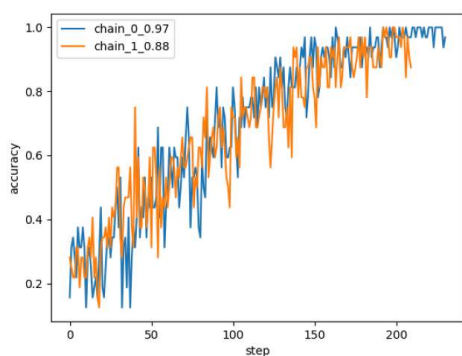
Total time: 39.8 min  
-----

## Warmstart Results

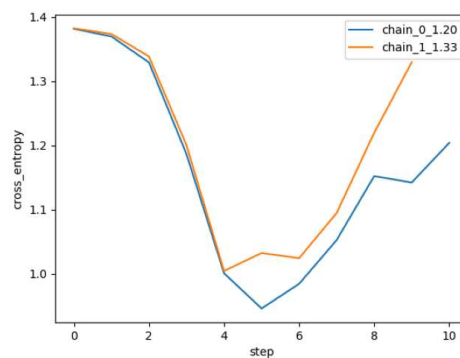
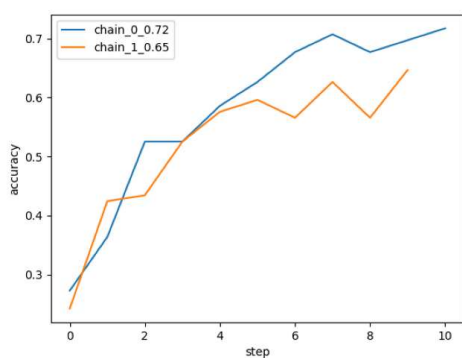
Number of NaNs:

```
{'MDPA': {'key': {'kernel': 0}, 'out': {'kernel': 0}, 'query': {'kernel': 0}, 'value': {'kernel': 0}}, 'TokenEmbedding_0': {'Embedding': {'embedding': 0}, 'PositionEmbedding': {'embedding': 0}}, 'classifier': {'kernel': 0}, 'projection_0': {'kernel': 0}}
```

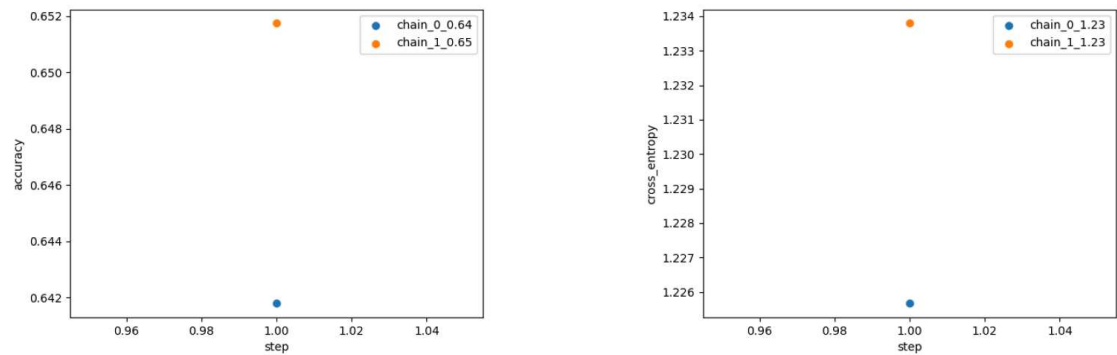
Train Set



Valid Set



Test Set



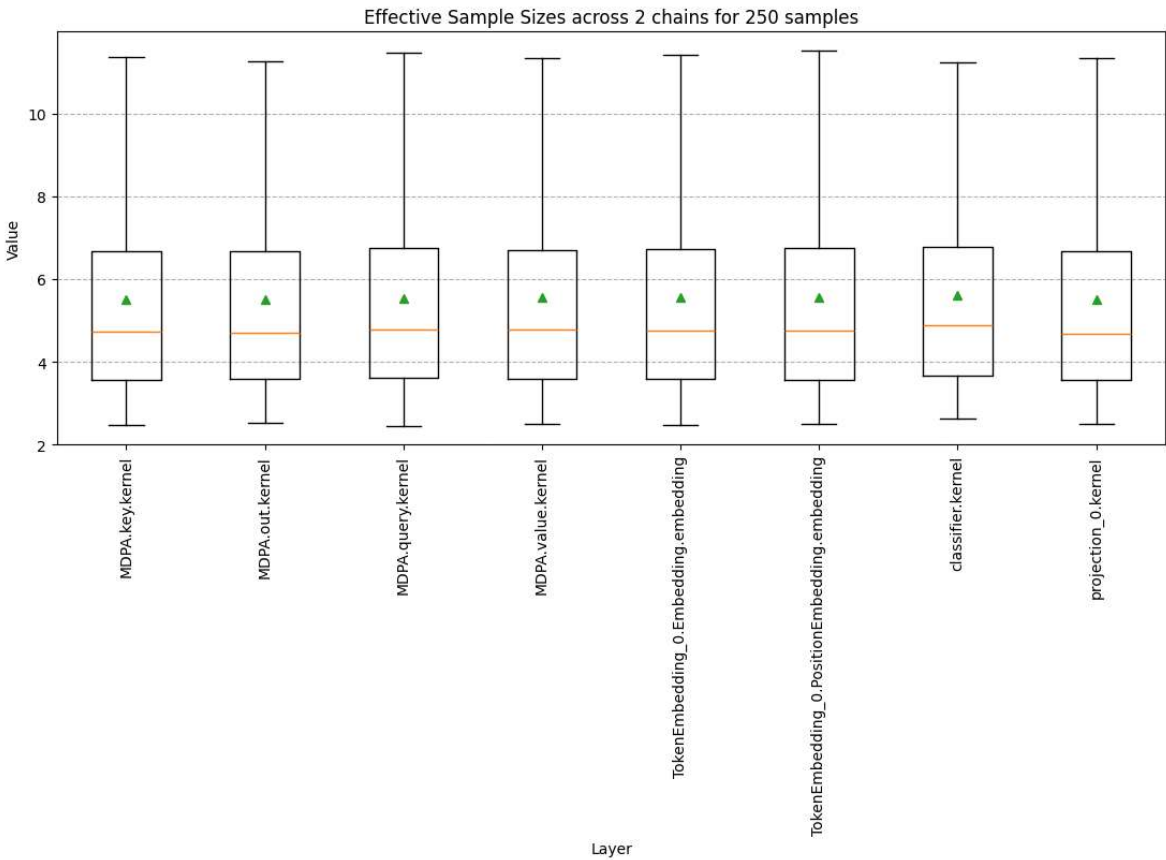
# Sampling Warmup

Tuned MCLMC Parameters after warmup:  
Step Size: 0.1847 ± 0.0000  
L: 1.0274 ± 0.0000

> No warmup samples has been saved.

# Sampling

Loaded 2 chains with 250 samples each



# Variance Analysis

```

-----
ValueError                                Traceback (most recent call last)
Cell In[9], line 2
      1 mean_bcv, mean_wcv = viz.plot_variances(samples)
----> 2 mean_crhat = viz.plot_split_chain_r_hat(samples, n_splits=4)
      3 # viz.plot_param_hist(samples, hist_limit=10000)

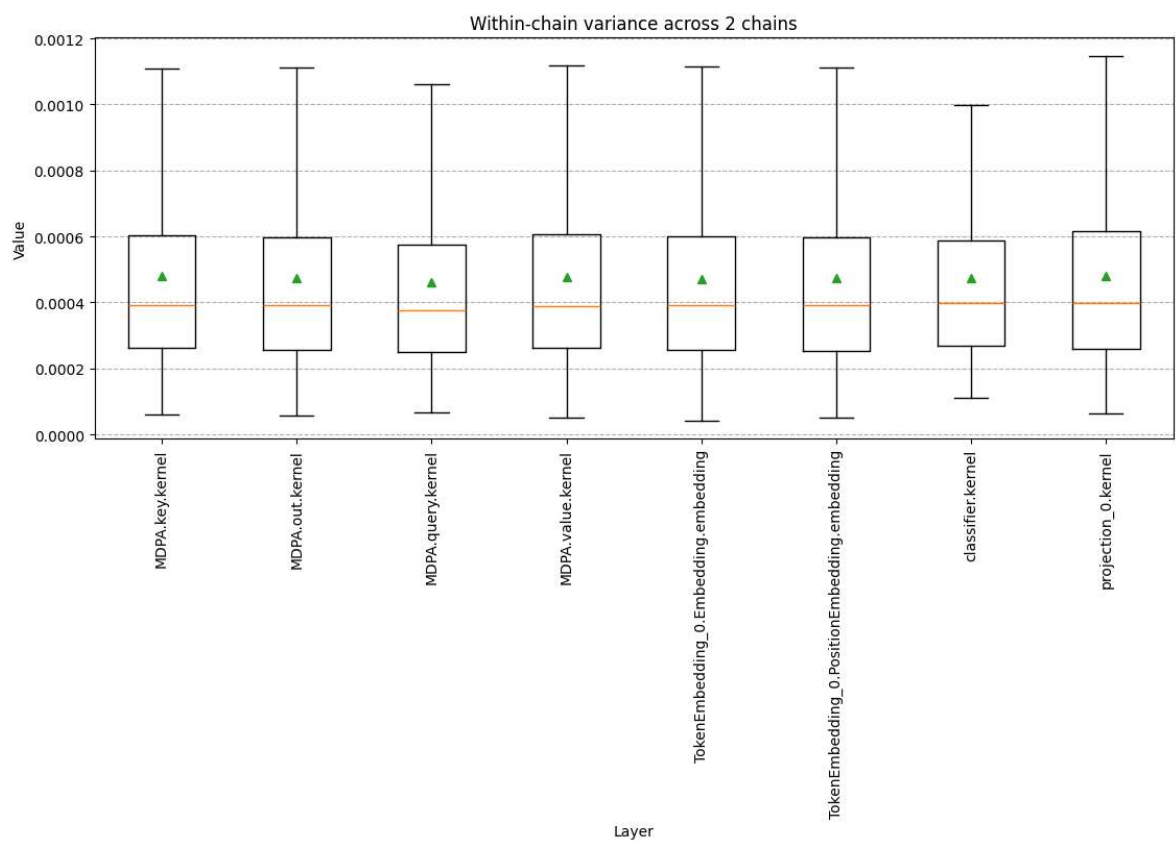
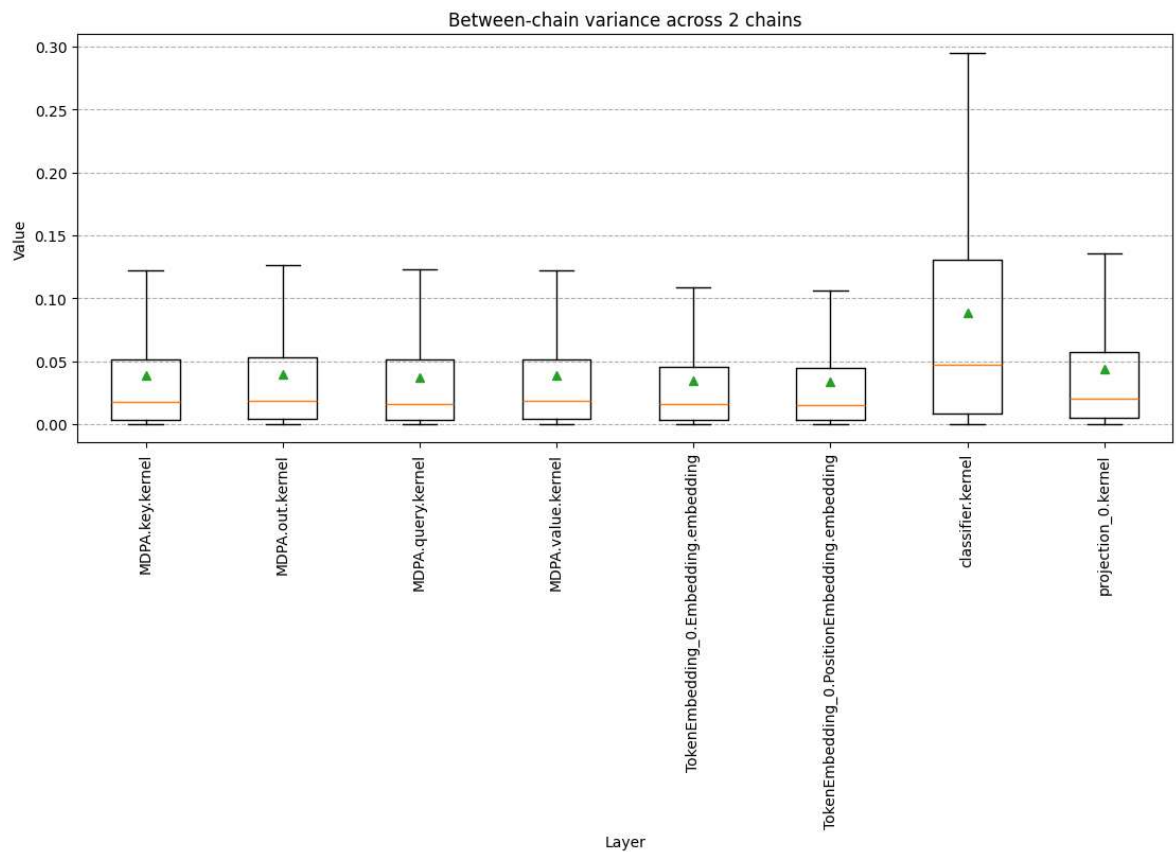
File /content/MILE/MILE/src/visualization/samples.py:205, in plot_split_chain_r_hat(samples, n_splits)
    202 n_chains = count_chains(samples)
    203 n_samples = count_samples(samples)
    204 rhat = {
--> 205     n: sandbox_metrics.split_chain_r_hat(v, n_splits)
    206     for n, v in zip(names, jax.tree.leaves(samples))
    207 }
    208 _plot_boxplots(
    209     rhat,
    210     f'{n_splits} Split Chain R-hat for {n_chains} chains each {n_samples}
samples',
    211 )
    212 return {n: jnp.mean(v) for n, v in rhat.items()}

File /content/MILE/MILE/src/inference/metrics.py:520, in split_chain_r_hat(samples, n_splits, rank_normalize)
    497 def split_chain_r_hat(
    498     samples: jnp.ndarray,
    499     n_splits: int,
    500     rank_normalize: bool = True,
    501 ) -> jnp.ndarray:
    502     """Calculate the split chain R-hat statistic for samples of MCMC chains.
ns.
    503
    504     Parameters
    (...
    516         Split R-hat statistic for each chain (n_chains, ...) as a JAX array.
ay.
    517     """
    518     return jnp.stack(
    519         [
--> 520             gelman_split_r_hat(chain[None, ...], n_splits, rank_normalize)
e)
    521         for chain in samples
    522     ]
    523 )

File /content/MILE/MILE/src/inference/metrics.py:474, in gelman_split_r_hat(samples, n_splits, rank_normalize)
    471 n_samples = samples.shape[1] / n_splits
    473 if (n_samples % 1) != 0:
--> 474     raise ValueError('Number of samples must be divisible by n_splits')
    476 if n_samples < 50:
    477     warnings.warn(
    478         message='Number of samples should be at least 50x the number of splits',
    479         category=UserWarning,
    480     )

ValueError: Number of samples must be divisible by n_splits

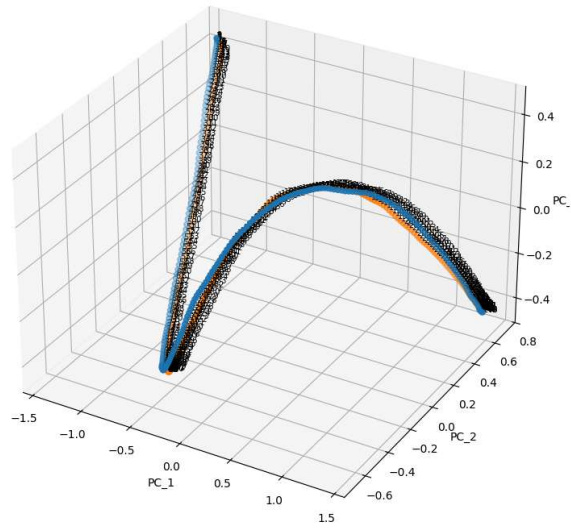
```



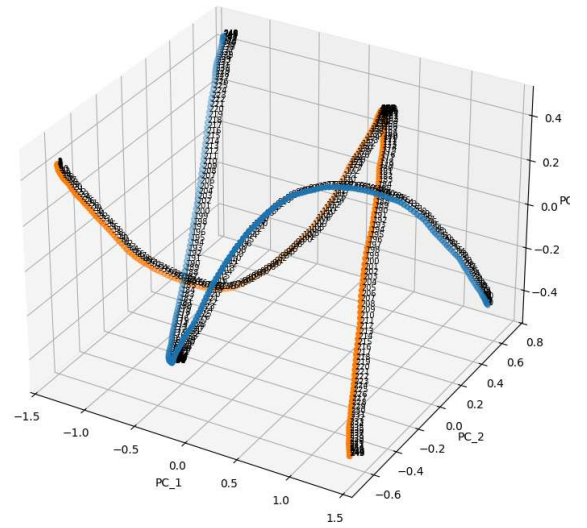
## Visualizing Sampling



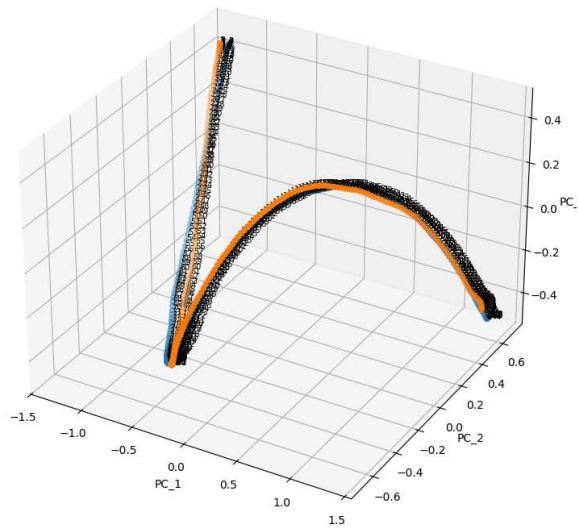
MDPA.key.kernel



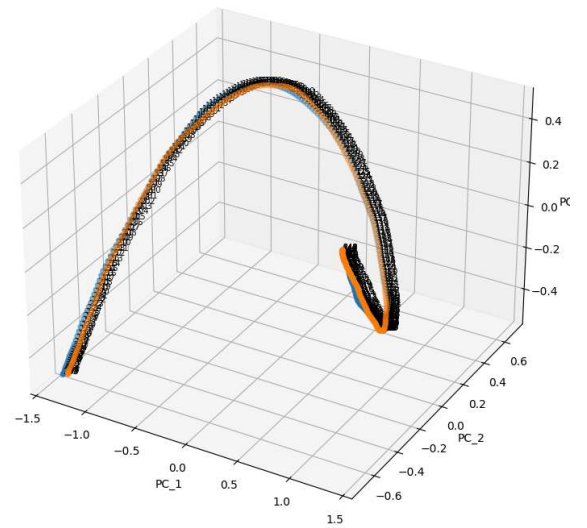
MDPA.out.kernel



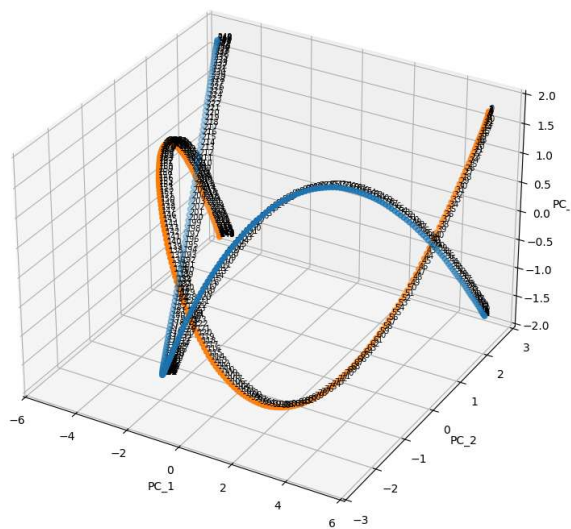
MDPA.query.kernel



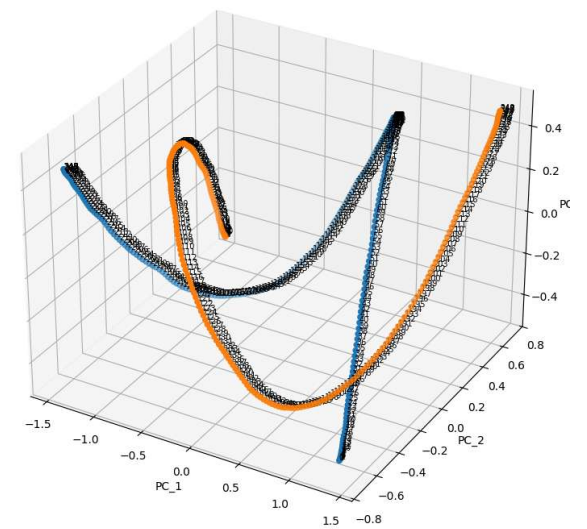
MDPA.value.kernel



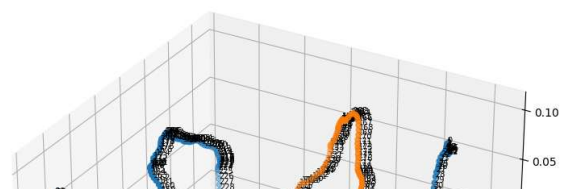
TokenEmbedding\_0.Embedding.embedding



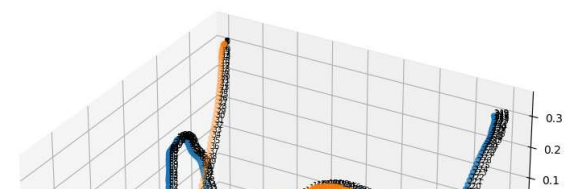
TokenEmbedding\_0.PositionEmbedding.embedding



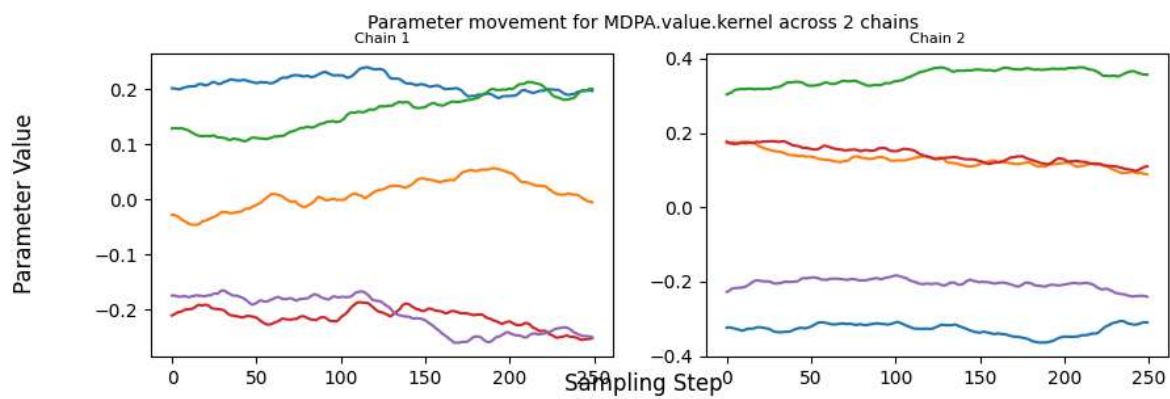
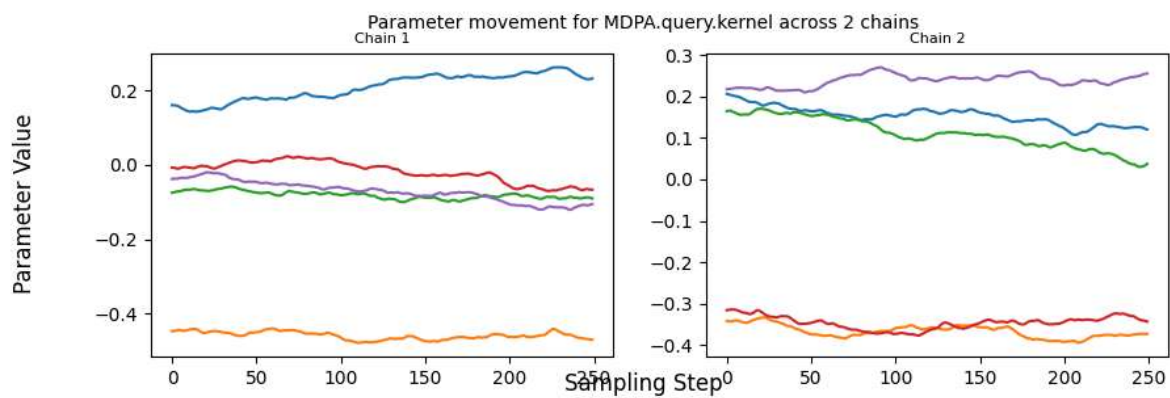
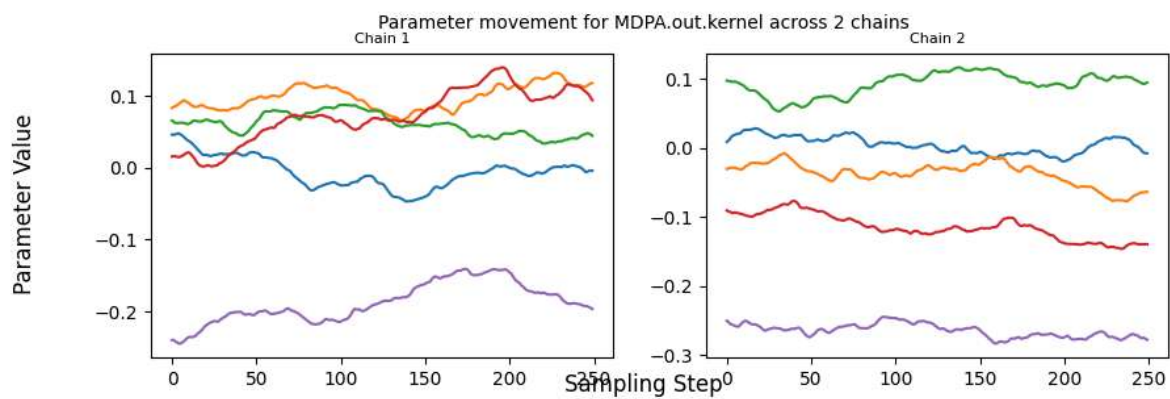
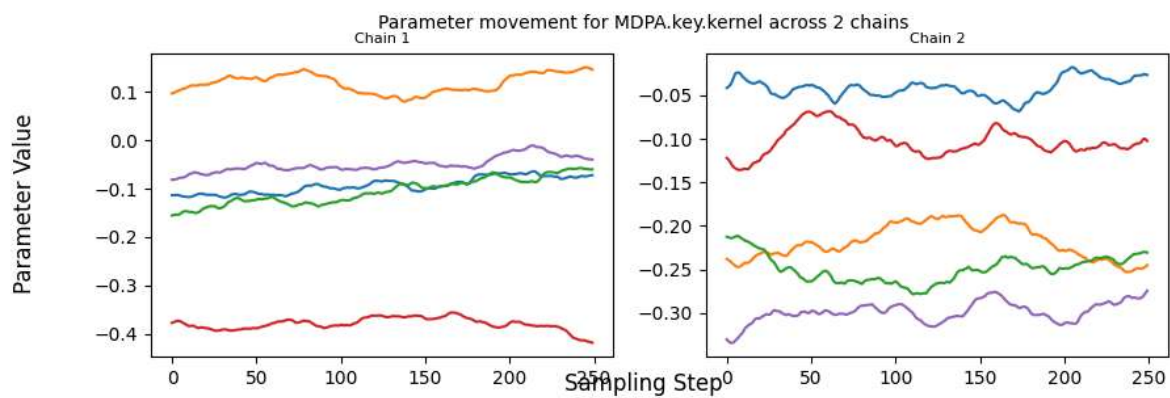
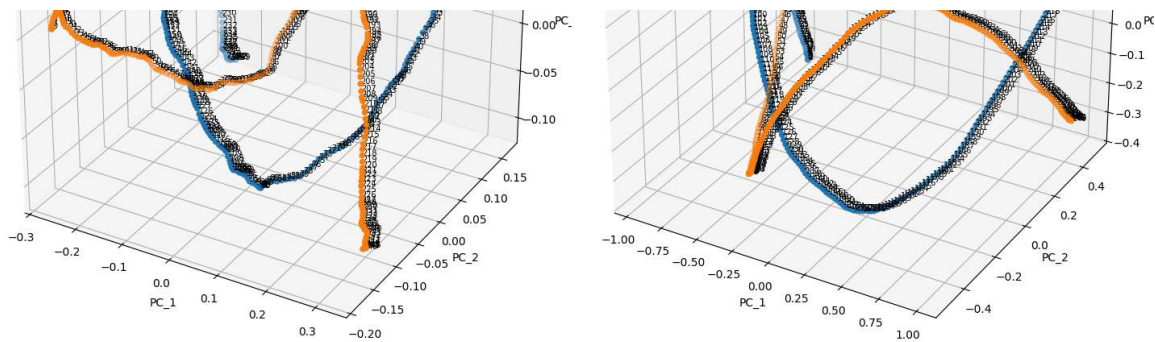
classifier.kernel

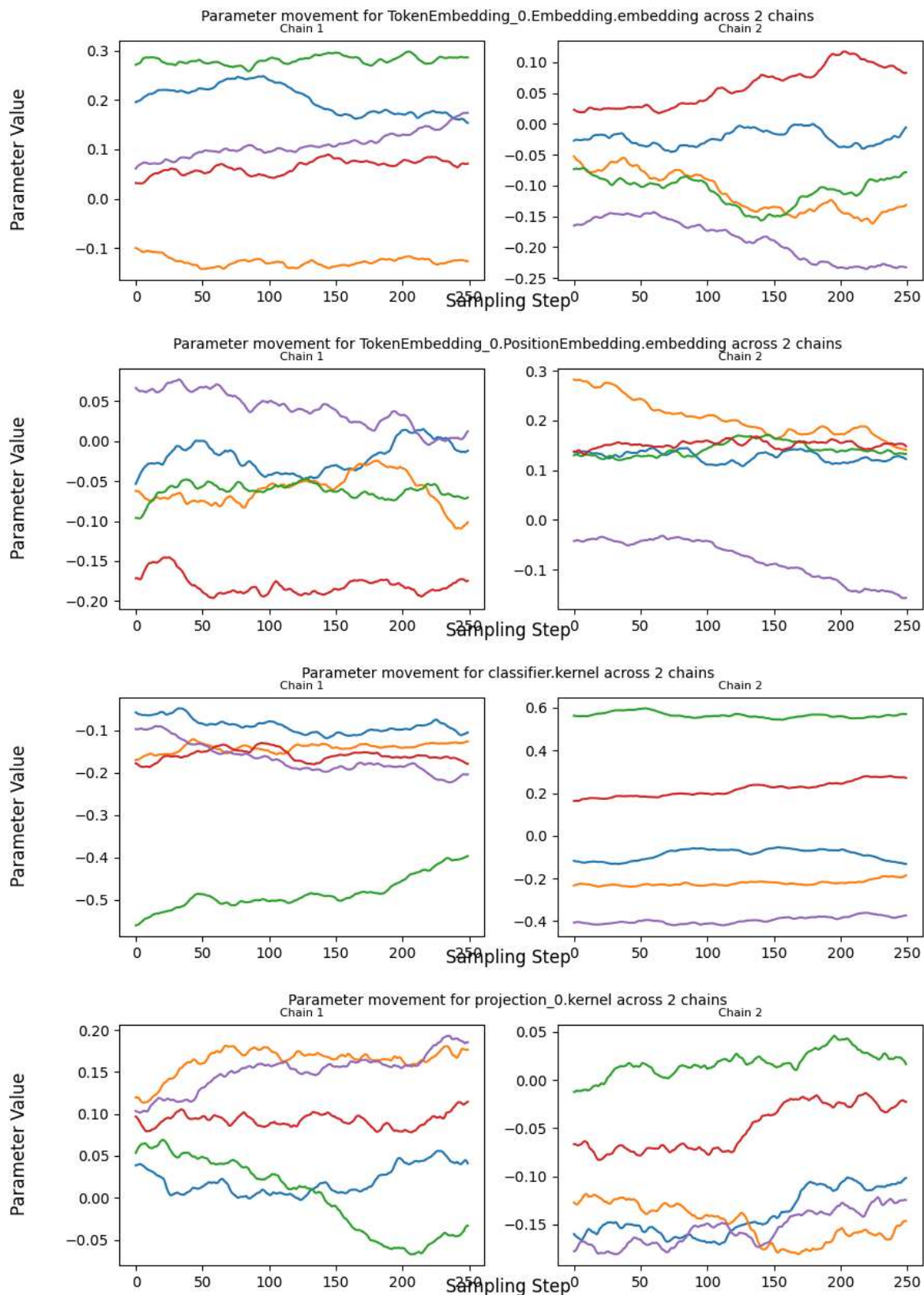


projection\_0.kernel









-----  
 NameError

Traceback (most recent call last)

Cell In[11], line 8

```

6 df['bcv'] = mean_bcv.values()
7 df['wcv'] = mean_wcv.values()
----> 8 df['crhat'] = mean_crhat.values()
9 df.to_csv(DIR / 'diagnostics.csv')
```

NameError: name 'mean\_crhat' is not defined



# DE vs BDE Performance

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_token.py:100: User
Warning:
Error while fetching `HF_TOKEN` secret value from your vault: 'Requesting secret
HF_TOKEN timed out. Secrets can only be fetched when running from the Colab UI.'.
You are not authenticated with the Hugging Face Hub in this notebook.
If the error persists, please let us know by opening an issue on GitHub (https://github.com/huggingface/huggingface\_hub/issues/new).
warnings.warn(
```

```

> Loader:
TextLoader:
| Data: ag_news
| Task: class
| Context Length: 70
| Vocabulary Size: 1000
| Target Length: 1
| Train: 700
| Valid: 99
| Test: 201
> Parameter Overview:

```

AttentionClassifier Summary

| path            | module          | inputs                                             | outputs         | params                                                                 |
|-----------------|-----------------|----------------------------------------------------|-----------------|------------------------------------------------------------------------|
|                 | AttentionCla... | pad_id: 0<br>train: False<br>x:<br>int32[1,70]     | float32[1,4]    |                                                                        |
| TokenEmbeddi... | TokenEmbeddi... | int32[1,70]                                        | float32[1,70... |                                                                        |
| TokenEmbeddi... | Embed           | int32[1,70]                                        | float32[1,70... | embedding:<br>float32[100...<br><br><b>48,000</b><br><b>(192.0 KB)</b> |
| TokenEmbeddi... | Embed           | int32[1,70]                                        | float32[1,70... | embedding:<br>float32[70,...<br><br><b>3,360 (13.4 KB)</b>             |
| MDPA            | MultiHeadDot... | -<br>float32[1,70...<br>- mask:<br>bool[1,1,70,... | float32[1,70... |                                                                        |
| MDPA/query      | DenseGeneral    | float32[1,70...                                    | float32[1,70... | kernel:<br>float32[48,...<br><br><b>3,072 (12.3 KB)</b>                |
| MDPA/key        | DenseGeneral    | float32[1,70...                                    | float32[1,70... | kernel:<br>float32[48,...<br><br><b>3,072 (12.3 KB)</b>                |
| MDPA/value      | DenseGeneral    | float32[1,70...                                    | float32[1,70... | kernel:<br>float32[48,...<br><br><b>3,072 (12.3 KB)</b>                |
| MDPA/out        | DenseGeneral    | float32[1,70...                                    | float32[1,70... | kernel:                                                                |

|              |       |               |               |                                                    |
|--------------|-------|---------------|---------------|----------------------------------------------------|
|              |       |               |               | float32[8,8...<br><b>3,072 (12.3 KB)</b>           |
| projection_0 | Dense | float32[1,48] | float32[1,32] | kernel:<br>float32[48,...<br><b>1,536 (6.1 KB)</b> |
| classifier   | Dense | float32[1,32] | float32[1,4]  | kernel:<br>float32[32,...<br><b>128 (512 B)</b>    |
|              |       |               | <b>Total</b>  | <b>65,312 (261.2 KB)</b>                           |

**Total Parameters: 65,312 (261.2 KB)**

Test Set: Feature and Label have shapes: (201, 70) (201,)

Deep Ensemble Performance | LPPD: -1.000, ACC: 0.6567

---

Chain 0 | LPPD: -1.226, ACC: 0.6418

Chain 1 | LPPD: -1.234, ACC: 0.6517

> Batch size for evaluation: 55.0

Calculating coverage is not supported for classification yet.

Bayesian Deep Ensemble Performance | LPPD: -1.021, ACC: 0.6169

---

Chain 0 | LPPD: -1.060, ACC: 0.5473

Chain 1 | LPPD: -1.093, ACC: 0.5721

